

Siddharth Balraj

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EDUCATION

Georgia Institute of Technology, College of Engineering, Atlanta, GA – GPA: 3.95/4

Master of Science, Aerospace Engineering (BS/MS Honors Program)

Starting Jan 2027

Bachelor of Science, Aerospace Engineering

Expected Dec 2026

SKILLS

Systems Engineering & MBSE: SysML v1/v2, CATIA Magic (Cameo, MSoSA), Teamwork Cloud

Simulation & CFD: STAR-CCM+, ANSYS (CFX, Fluent), Cadence Fidelity, Pointwise, Tecplot, ParaView, NASA CEA

Aircraft Design & CAD: AVL, OpenVSP, XFOIL, XFLR5, SolidWorks, Creo, Visio

Programming: Python, MATLAB, Simulink, VBA, Java, TypeScript, Linux (HPC), Excel

EXPERIENCE

Undergraduate Research Assistant, Aerospace Systems Design Laboratory (ASDL) Jan – May 2025, Jan 2026 – Present

- Built a complete, executable MBSE model in SysML v2 (CATIA Magic) to design and verify an automated aircraft landing flare control system, unifying requirements, architecture, behavior, and analysis in one model.
- Designed an LQI flare controller on a linearized 4-state longitudinal dynamics model in MATLAB/Simulink, using an altitude-scheduled sink-rate reference to **satisfy all six touchdown safety and performance requirements**.
- Integrated SysML v2 with MATLAB for one-click closed-loop simulation and automatic pass/fail requirement verification; **validated instances across a 140:1 aircraft mass range** with automated failure detection.
- Modeling how AI factory growth loads regional power and water infrastructure on a graduate research team with Siemens; planning the system-of-systems architecture and modeling frameworks to test siting, staging, and cooling trades.

Propulsion Engineering Intern, SPARC Research

May – Aug 2026

- Built a 2D axisymmetric solid fuel ramjet (SFRJ) combustor model in STAR-CCM+ with combustion chemistry from published reduced kinetics, achieving a stable reacting-flow solution.
- Developed and automated a burnback method for complex 3D propellant grains in SolidWorks (VBA) and STAR-CCM+ (Java); validated against closed-form and legacy results, **cutting runtime from ~1 day to minutes at higher resolution**.
- Automated selection of the fuel-grain-adjacent cell layer for pyrolysis source terms with a self-calibrating STAR-CCM+ Java macro, **replacing an eight-step workflow** and validated against compiled C user code.
- Developed a Python tool porting NASA CEA 9-coefficient thermodynamic polynomials into the STAR-CCM+ material database, correcting erroneous default species data used in SFRJ combustion simulations.
- Wrote user documentation for the burnback tools and presented the completed work to the full ~35-person company.

Aerodynamics Engineering Intern, Trane Technologies (full-time summer 2025, continuing part-time) May 2025 – Present

- Generate compressor operating maps and minimum-capacity plots for single- and two-stage centrifugal designs using Trane's steady-state solver, Python, and Tecplot; present results to project leadership as justification for design selections.
- Simulated 8 centrifugal compressor configurations with full-stage CFD in Ansys CFX using high-fidelity meshes built in Cadence Fidelity, identifying a configuration with **2.4% higher stage efficiency than the production baseline**.
- Evaluated angled vs. radial IGV motor-cooling injection methods for flow uniformity and pressure losses; the radial design was adopted for easier manufacturability at only a 0.1% stage-efficiency penalty.
- Automated multi-configuration CFD runs and post-processing through Linux HPC workflows and refactored internal Python data-reduction scripts, **cutting code length by 27%** and accelerating turnaround for test engineers.

Systems & Mechanical Engineering Intern, GE Vernova

Sept – Dec 2025

- Ran open-enclosure airflow and conjugate heat-transfer simulations for solid-state transformer design iterations; designed internal air-duct guides that **cut coil temperature non-uniformity from ~50°C to 16°C**.
- Implemented pressure-flow fan-curve boundary conditions in ANSYS Fluent to evaluate fans from multiple manufacturers, identifying ventilation combinations that **reduced core temperature by 24°C** relative to baseline.
- Created 2D/3D CAD models of switchboards and load centers for two multi-million-dollar naval variable-frequency-drive proposals; presented completed work to engineering leadership.

PROJECTS

Collaborative Combat Aircraft (CCA) Design – Senior Aircraft Design II

Fall 2026

- Designing an unmanned, attritable CCA with a student team to a Mach 0.9, 600 nm combat radius, 2,000 lb internal payload specification; built the Class I weight regression from 14 comparable CCAs, UCAVs, and light attack jets and am evaluating six configurations modeled in OpenVSP against weighted figures of merit.

Middle-of-the-Market Commercial Airliner Design – Senior Aircraft Design I

Spring 2026

- Completed the conceptual design cycle for a 280-passenger, Mach 0.80 airliner: configuration selection, weight and constraint sizing, cost estimation, XFLR5 airfoil selection, AVL wing analysis, and OpenVSP fuselage and tail sizing.